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The “humane in the loop”: Inclusive research design and policy approaches to foster capacity building assistive technologies in the COVID-19 era

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ABSTRACT

The COVID-19 pandemic is emerging as a driver of greater reliance on wireless technologies, including intelligent assistive technologies, such as robots and artificial intelligence. We must integrate the humane “into the loop” of human-AT interactions to realize the full potential of wireless inclusion for people with disabilities and older adults. Embedding ethics into these new technologies is critical and requires a co-design approach, with end users participating throughout. Developing humane AT begins with a participatory, user-centered design embedded in an iterative co-creation process, and guided by an ethos prioritizing beneficence, user autonomy and agency. To gain insight into plausible AT development pathways (“futures”), we use scenario planning as a tool to articulate themes in the research literature. Four plausible scenarios are developed and compared to identify one as a desired “humane” future for AT development. Policy and practice recommendations derived from this scenario, and their implications for the role of AT in the advancement of human potential are explored.

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Intelligent Assistive Technologies; Inclusive Design; Ethics; Disability; Artificial Intelligence

Background

The COVID-19 pandemic has emerged as a driver of greater reliance on wireless technology including robots, Artificial Intelligence (AI), big data analytics, mobile apps, social media, online education, and telehealth (Mazzoleni et al., 2020; Tavakoli et al., 2020; Ting et al., 2020). Wireless connectivity and digital technologies have enabled a myriad of applications during the COVID-19 pandemic, including flexible working environments, virtual reality-based training, and distance education that are accelerating the digital transformation of public life (Butt et al., 2020). Innovative technologies, partially in response to the pandemic, have moved us increasingly toward mediated communications, experience, and exploration (Papadopoulos et al., 2020; Tavakoli et al., 2020). The accelerated use of technology in response to COVID-19 may point to an inflection point for future technology applications in health care and beyond (Garfin 2020; Tavakoli et al., 2020), building on trends in assistive technologies (Thordardottir, 2019), and what has been described as a “smart revolution” (Kosta et al., 2018). This trend is not unproblematic and, in fact, brings with its ethical dilemmas around possible discrimination, privacy violations, and bias, including particular concerns for people with disabilities and older adults (Kosta et al., 2018).

In our analysis, we focus on assistive technologies, in particular, intelligent assistive technologies (IAT), inclusive of robotics, devices such as tablets, wearables, voice activated assistants, distributive systems such as smart homes, integrated sensor systems, mobile platforms, and artificial intelligence

(Bharucha, et al., 2009; Dahlke & Ory, 2020; Ienca et al., 2017; Wangmo et al., 2019). Intelligent assistive technologies are distinct from other assistive technologies in that they possess computational capabilities and, in contrast to mechanistic tools, are capable of adapting to changing situations while being responsive to their user’s needs and the environment (Bharucha, et al., 2009; Ienca et al., 2017; Wangmo et al., 2019). Intelligent assistive technologies enhance the capabilities of users, including people with cognitive or physical disabilities, to perform needed activities of daily living, engage socially, and live independently longer than they might otherwise (Wangmo et al., 2019). They also provide care and assistance, companionship, social and emotional support (Ienca et al., 2017).

The design and development of ethical intelligent assistive technologies (IAT) in the context of the COVID-19 era has profound implications for their impact on the lives of older adults and people with disabilities as we look toward an emerging “new normal” of social, economic, and political life. There are a number of ethical issues surrounding the use of IAT, owing to their sophistication (e.g., vulnerability to bias), sensing, data collection, and communication abilities (e.g., risks to privacy), as well as the assistive roles they play close proximity to users (e.g., risks to harm). These users are vulnerable by dint of their disability, generating ethical issues around preventing harm, safeguarding privacy, meaningful consent, and limited design inputs. Similarly, constraints on accessing the technology, due to expense, usability, and digital divide challenges present ethical issues that need to be resolved. While there is a subset of IATs that are available directly to the consumer

(some smartphones and embedded GPS), most IATs are costly, both to develop and purchase. They are only cost-effective by comparison to the alternative (loss of independent living and functioning), thus introducing affordability and access equity issues (Dalhke & Ory, 2020; Wangmo et al., 2019). The larger class of devices and systems for which IATs are a part – artificial intelligence and persuasive and ubiquitous computing operating in a wireless environment – function as IAT in care settings and, more broadly, environments where human-computer interactions are aimed at supporting the needs and aspirations of people with disabilities and other vulnerable populations (Ienca et al., 2017). The design and development of devices and systems serving as IATs must then anticipate and respond to these ethical concerns, requiring trade-offs. For example, trade-offs between optimization, for example, prioritizing tracking or data collection accuracy, and designs that prioritize beneficence in terms of assistance and capability-building, or safeguard against harms and malfeasance, sacrificing precision (Ienca et al., 2018).

Robots that respond to, and interact with the environment in a non-repetitive manner are said to be either semi-autonomous or autonomous, and exercise some degree of agency in action and decision-making. Robots that provide some measure of aid to and interaction with people meet the criteria for being artificially intelligence (AI)-driven: they receive, process, and stream information to plan and complete tasks (Shyr et al., 2018). Service robots and other robots functioning as AI, in contrast to robots that perform repetitive tasks in industrial settings, generally learn to use a scripted approach, machine learning approaches, or automated planning techniques. The machine learning approach constitutes a “black box” regarding the algorithms antecedent to action, although some measure of transparency is possible insofar as properties of the training data set and the role of human facilitators or oversight may be known (Magrani 2017; Sunderhauf et al., 2018). The third approach, automated planning techniques, relies on partially or completely ordered action sequences that alter the observed environment to obtain a sought after goal state to inform high-level action (Capitanelli et al., 2017; Foster et al., 2020).

At present, robots are semi-autonomous and subject to human intervention in design and implementation, making it possible, at least in principle, to anticipate ethical and practical harms, while assigning responsibility for breaches. A social robot purposed to decrease social isolation for a user with a disability, but instead serves to replace human interaction, creates a harm that the human designers must anticipate and avoid by attending to both programming and context of use. Apparent ethical agency on the part of the robot stems from human involvement in their development and use. Robots remain amoral tools, at least until they can comprehend the normative and ethical legal, social, and cultural standards, with the onus of responsibility for the robot’s engagement with people and society on the designers, and human oversight (Westerlund, 2020). Policy will need to address how the linkage between AI and robots nonetheless leaves a human-centered gap in the ethical and legal responsibilities of robots as semi-autonomous entities. Their human programmers, and/or the data sets and models instituted by

humans to train them, bear significant responsibility for any harms the robots inflict.

As AI, robots and other intelligent AT are paired with humans to generate innovative solutions to the aspirations of people with disabilities and older adults, we must integrate the humane “into the loop” to realize the full potential of wireless inclusion (i.e., Leveringhaus, 2018; Torresen, 2018). Wireless inclusion engages participants, beyond simply granting them access and useable interfaces. In the absence of end-users in the design and development of human-machine collaborations, inclusion will remain an elusive goal.

A participatory ethos, grounded in the values of user autonomy and agency, must guide the design, implementation, and assessment of blossoming wireless assistive technologies in the COVID-19 era to realize inclusion and accessibility for people with disabilities and older adults. Individual autonomy is fundamental to choice and agency – having the capacity to take action, is critical to realizing the potential for autonomous choice (Scully, 2010; Wayne, 2019). Safety, security, human contact, and caring relationships also support individual agency and are key components of ethical assistive technologies (Mansouri et al., 2017; Zwijsen et al., 2011). The bar for “humane” in the loop is a high one, not merely including a human voice in the development of technology, but also extending to compassionate caring for human welfare (Hugman, 2012).

Intelligent systems, composed of robots, wearables, intelligent assistants, and other intelligent agents are anticipated to coordinate and collaborate with humans in what has been termed “hybrid” or “extended” intelligence (Peeters et al., 2020). This movement toward integrated human-machine intelligence and decision-making holds the potential to assist individuals with a variety of disabilities, for instance, people with physical or cognitive disabilities navigating physical space to travel to work, shop, recreate, or socialize. A combination of networked devices, including social robots, could help them map out and strategize pathways, while providing physical assistance and/or decision-making aids in response to the user’s preferences and goals. In this example, an individual with a cognitive disability could be assisted by the social robot in practicing how to seek directions, if lost, as well as receiving assistance visualizing, planning, and rehearsing the travel process. The social robot would be the node on the machine side of the equation, with the user, the locus of the human-agent side. The robot’s actions would be aimed at maximizing the benefits to the user while minimizing the costs. Beneficence and ethical safeguards for the user’s rights and well-being will serve as the heuristic for gauging the efficacy of a given action, rather than optimization of effort, speed, or other metrics. One ethical risk associated with the use of a social robot for enhancing the social skills of a user with an intellectual or developmental disability (IDD), for example, is that the robot would come to substitute for a human connection, either because the user is deceived about the humanity of the robot, or because potential human relationships are withdrawn, or not made available.

We examine the role of user-centered, participatory design approaches in realizing the “humane in the loop.” We then use a scenario planning approach to explore future possibilities in

these uncertain times, guided by the relevant research literature, to identify and assess four possible pathways for the development of ethical, humane intelligent assistive technologies. The scenario planning approach allows us to obtain insights into which scenario is most likely to best foster humane outcomes and processes in the loop of human-AT interactions. After examining each of the four scenarios (“Moral agent”, “programmed”, “human liability, and regulated”), we identify one scenario with the greatest potential to infuse humane priorities into policy and practice recommendations for ethical AT development. Subsequently, we explore the implications of humane and ethical AT for the advancement of human capabilities.

Methods/materials

Problem formulation

Wireless inclusive assistive technologies (i.e., IAT), such as AI and robotics simultaneously hold the potential for isolation *and* engagement, enhanced capabilities *and* restrictions, greater knowledge *and* misinformation. How the technologies are developed, guided by design decisions at every step of the process, is critical to fashioning AT that expand human agency and autonomy as ethical imperatives.

Design thinking reflects the needs of users and informs both infrastructure and systems (Brown & Wyatt, 2010). Its power lies in the overlapping drives of inspiration, ideation, and implementation and provides innovative, bottom-up, systemic solutions to systemic problems (Brown & Wyatt, 2010). Design thinking incorporates multiple perspectives, iteratively, engaging stakeholders early in the process, discovering the users’ needs and generating alternative solutions prototyping and testing (Brown & Wyatt, 2010; Minton & Luetjens, 2016). User centered, or inclusive design generates maximally accessible interfaces, artifacts, products, and services guided by the user’s problem definition and inputs (Reed & Monk, 2011; Wilkinson & De Angeli, 2014). Considering the use-context, including social, cultural, and economic factors, is key to user-centered design to help ensure that design does not become a bolt-on artifact, but rather serves as an integral part of the development process (Gandy et al., 2017). A design approach to developing wireless technologies bids that we actively engage users (people with disabilities and older adults) in a process that prioritizes their desires and aspirations. Such aspirations can run the gamut from esthetics to efficiency, as appropriate to the use case. Rapid prototyping and a co-creation process characterize design-led innovation, thereby allowing for corrective feedback and adaptations (Fleischmann, 2019). As part of complex socio-technological systems, wireless technologies, such as IAT, evolve in a non-linear fashion. Thus, the problem formulation process must be tuned to non-linear analytic methods for framing and solving social problems (Martin et al., 2020). Anticipating potential trajectories for wireless technology development that are non-linear and probabilistic, we turn to foresight methods and futures-oriented thinking, which anticipate new technology trends and advancements (Hussain et al., 2017; Wiebe et al., 2018). These approaches are suitable for flexible policy-making decisions, informed by

plausible and stakeholder-informed pathways for technology development (i.e., Baker et al., 2015; Gandy et al., 2017).

Marginalized populations, including people with disabilities and older adults, are too frequently left out of the design loop. A participatory design process is the antidote to the neglect that the stories, aspirations, and needs IAT users often face (Baker et al., 2015; Gandy et al., 2017). Design that has a future orientation aims to re-envision how new technology is creative, exploratory, and draws inspiration from possibilities (Micheli et al., 2019). The costs associated with developing new IAT, while an impediment, on the one hand, can also be a spur to innovation on the other, as low-cost alternatives are developed (Ienca et al., 2018). Participatory design extends the user-centric premise further, in a co-creation process with ideas and insights drawn from all people at every step of the design process (Wilkinson & De Angeli, 2014). User-centric design enhances the IAT user’s autonomy, dignity, and capabilities, and is a beneficence-oriented, value-sensitive approach to resolving competing interests empowering the end user (Ienca et al., 2017, 2018). To realize the “humane in the loop,” a participatory, user-centric design process is ideal, and serves as a bridge to imagined futures in which intelligent assistive technologies and supportive policies are developed that resonate with the needs and aspirations of people with disabilities and older adults.

How the intelligent agents are trained has strong implications for both ethical and technological considerations of AI and robot performance, especially around transparency. Transparency is key to both ensuring ethical safeguards and holding machine performance accountable for any harms. Machine learning, which produces algorithms with minimal human input, and even more so, deep learning, for which the algorithms truly constitute a black box, begs the question of how the machine/robot learning is instantiated through the training data set and model. Ethical issues arise around where those data are from, how they were collected, what they were meant to address, and what is missing from the data set, in addition to any biases: systematic, random, known, or unknown, that might lurk. Being able to answer at least some of these questions, while not wholly raising the veil from the black box, will at least provide some measure of transparency. In order to make the algorithms more understandable to humans requires scripted or automated planning approaches that exert more human control and specification, and are thus more readily traceable and comprehensible.

Guided by the principle that robots and AI ought to be optimally responsive to human needs, goals, and human guidance, interfaces must aim at being more iterative, flexible, and customizable (McGinn, 2020). This is particularly true in social contexts, for which socially assistive robots are indicated. In social robot-human scenarios, the complexity of social interactions militates against simple predictions and requires a high degree of adaptability at the interface, together with ongoing evaluation of the interactions and incremental modifications (McGinn, 2020). Robot social interfaces represent a promising approach for human-centric design, and are amenable to participatory design, either stand alone or in concert with other intelligent devices (McGinn, 2020). A use case example might involve a social robot providing assistance to an adult with

a cognitive disability navigating the complex systems of a smart home, and different scenarios, such as accessing entertainment, emailing, or altering lighting for guests. Having a choice of interfaces to choose from according to the user's preferences might include a social interface provided by the robot, voice interfaces with intelligent devices, or screen controls, depending upon the use context, with the user's preferences captured through human-centered designs (Shiffhauer et al., 2016).

There are a number of ways to train an AI to be ethical, including supervised machine learning algorithms, machine learning, or deep learning with big data (Garcia, et al., 2020; Jenkins, et al., 2020). Consider a use case for training a socially assistive robot, Pepper. In this case, Pepper engages a young adult care recipient with intellectual or developmental disabilities (IDD) in a learning scenario to provide the older adult with respite. Pepper's social interface (sensors) can be tasked with monitoring that is more or less invasive of the young adult's privacy to ensure that the respite scenario is both safe and effective. This will include considerations of what is ethical to share with the caregiver in real time and/or after the scenario. One training approach might use scripted algorithms derived from a participatory research design. The normative standards for ethical caregiving will be derived from the caregivers' and care recipients' interviews, and combined with inputs from the design team, who are trained in the ethical fundamentals. These data will be used to train Pepper, making corrections accordingly. The community responsible for the training in this instance engages the robot in a context-dependent learning process bearing the stamp of the individuals involved, analogous to child-rearing as we do it in "our family". A contrasting approach uses machine learning, grounded in baseline data on models of safe and privacy-enhancing behaviors, against which the robot's behaviors are evaluated. The robot's performance is assessed on the likelihood of the young adult's privacy, or safety, being put at risk (and corrected accordingly). This is akin to child rearing in accordance with more generic, population-based principles. A learning philosophy like this can be applied to multiple data sources efficiently, and contrasts with one that is founded in structured training and more nearly resembles hands-on child training, rather than prediction against a model. These choices can be at least partially explained by the values and priorities of the community that designs AI learning; whether the focus is on more effective technological solutions (i.e., greater efficiency) or if it privileges socio-cultural solutions (i.e., situated ethical standards).

Futures perspectives

Strategic decision-making in the COVID-19 era of uncertainty and rapid change requires insights into plausible futures, for which scenario planning is a foresight technique that facilitates a flexible, multi-perspective approach toward envisioning a desired future (Meyerowitz et al., 2018). As technology evolves, there is a recognition of the value of tools that enable foresight, rather than predictions, recognizing the indeterminate nature of the "future". Hence, the alternative term "futures" is used. Scenario planning is a four-stage process of articulating a likely "scene", identifying then ranking key

driving forces, and generating plausible scenarios around emergent themes can serve as a powerful technology foresight tool (Hussain et al., 2017).

When planning for a more desired future, we must first confront its essential unknowable nature. That is, there are any number of possible futures, and the one that emerges may or may not correspond to our expectations. This suggests that in addition to using foresight methods to explore possible futures, we also need to institute a rigorous design and evaluation regimen that will allow course corrections as the situation changes, different scenarios are tested, and new data come in. The future is not only in the making, it is also in the re-making, empowering the users to provide feedback on the desirability of the scenarios, and supporting the evolution of the machine-human collaboration. Robot/AI contributions to greater future capabilities, accessibility, and participation for people with disabilities, as a marginalized and vulnerable population, are predicated upon the evolution of more transparent, accountable, and empowering human-machine interactions and collaborative scenarios.

We employed a scenario planning approach to plotting out possible pathways for the development of intelligent AT (specifically robots and AI) imputing a time horizon is of five to ten years, taking into account the pace of technological development. Ethical questions around the design, implementation, and the consequences of IAT performances are critical to assessing their impact. The ethical character of machine behavior is undergirded by the presumed (or inferred) moral capacities of the machine – these are intertwined, but distinct concepts (Malle, 2016). More recently, Cervantes and colleagues distinguish between the underlying moral dimensions of human behavior (that which is good, just, and proper) and artificial ethical agents, exhibiting behaviors, whether based on ethical rules, or as part of their functionality (Cervantes et al., 2020).

We generated four plausible alternative scenarios (story lines) guided by the extant research literature on ethics, morality, robotics, and AI. Our literature review was qualitative and "semi-systematic" that is, drawing upon a complex topic (machine ethics and morality) that conceptualized in various ways, making an exhaustive review impossible, but from which we could detect broad themes (Snyder, 2019). Our foresight process was guided by scenario planning literature (Burt & Nair, 2020; Lexa & Chan, 2010; Lustig, 2017; Moats et al., 2008; Ralston, 2008). The scenarios spur insights as they aim not at discrete predictions, but rather at generating insights useful for strategic thinking and learning (Edgar et al., 2013). After exploring each scenario, we assessed its likely probability and impact. On that basis, we chose one with the highest likelihood of facilitating ethical technologies. The process identified a desired future – one in which the developmental pathway of IAT is most likely to enhance the autonomy and agency of older adults and people with disabilities. That scenario will be provided the basis from which we will make policy and practice recommendations.

Results

In the *Moral Agent* scenario, the robot or AI operates as a moral agent; algorithms created through machine learning

govern its behavior. The machine uses data – for robots, this includes data from human interactions – to make predictions that direct its performance (Cuayahuitl et al., 2014). In this scenario, the robot or AI is the moral equivalent of a human, having the same rights and responsibilities, even though the machine's decision-making process is a black box, and it need not parallel that of a human. However, in this context, the robot or AI is the functional equivalent of a person, and it could quite possibly have the legal standing of a human. With some reservations, it could, in principle, be a moral agent, or at least morally competent (Gunkel, 2012; Malle & Scheutz, 2015; Sommer et al., 2019). This is despite certain key limitations, namely, that robots and/or AI have yet to reach this threshold; machine learning is vulnerable to systematic biases, and the appearance of morality does not signal its substance (Sharkey, 2020; van Wynsberghe & Robbins, 2019). This scenario is quite futuristic and out of reach, with a long-time horizon, as autonomous, moral robots, and/or AI are not likely to be operating anytime soon (Cervantes et al., 2020).

The *Programmed* scenario posits that the robot or AI is in itself (*sui generis*) neutral from a moral or ethical perspective – a machine into which ethical rules are programmed, and “moral” behavior, from the human perspective, is the result (Vandemeulebroucke et al., 2020). What observers may deem morality or ethical behavior on the part of AI or robotics is in fact an epiphenomenon – moral patency rather than true agency is at play (Gunkel, 2012).

As we envision it, in this scenario, the technology is truly value-neutral and any behaviors of perceived ethical consequence arise from the misapprehension of the robot or AI as more than a tool or device, however complex or computationally advanced. It is ontologically, if not functionally akin to, say, a car airbag, which deploys automatically to preserve the life of the passenger with minimal situational “awareness” and no sense of what is at stake. Averting harm, or providing safety for its user, is a simple function of what it was designed to do, granted that the design is far more sophisticated and complicated than that of an airbag release system. In this scenario, human programming of robots is the source of robot behavior. This particular scenario is less likely to emerge ascendant than the other scenarios, in part because machine learning can mine the insights of multiple interactions, rich data sets, and distributed networks, thus producing superior results. While intensive programming by humans will likely remain a component of ethical robot and/or AI performance, the Programmed scenario presents some limitations in comparison with that of other scenarios.

In the *Human Liability* scenario, the robot or AI has no accountability for any harms arising from human-machine interactions, or machine behaviors. In this scenario, the technology is a human-made artifact, and it does not have partial or joint moral responsibility. Moral responsibility rests entirely with the human or humans (Hakli & Makela, 2019). We envision humans regulating the ethical behavior of robots and/or AI in this scenario by selectively using a “kill switch,” turning off the machine to prevent it from doing harm. The bottom line for users is to follow the user manual because operating outside those parameters is to assume the risk. It is difficult to conceive of this scenario as one that will gain much traction in the future.

The notion of value-neutral, “objective” technology derived from machine learning has been invalidated with notable cases, such as Microsoft's “Tay” chatbots “tutored” in racism from Twitter data. Machine learning-based interventions and tools concomitant with the development of AI and robotics bring greater predictive power and objectivity, broadly speaking, but they can also propagate unfair outcomes to marginalized communities because of faulty socio-technical problem formulation based on biased (human) causal theories and inferences (Martin et al., 2020). The iterative exchanges between human and machine that bias the machine's algorithms and skew its performance may be a result of how people selectively label information (through a personal lens, randomly or through active learning), and the algorithm's own process of information selection (Sun et al., 2020). Holding the machines harmless is to put too little emphasis on the consequences of their nonobjective learning process.

In the *Regulated* scenario, robots and AI are at least partially accountable for their actions as semi-autonomous actors. The argument is that because these technologies are capable of modulating peoples' behavior, AI, and robots are partially responsible for the consequences (Hakli & Makela, 2019; Kim & Kim, 2013). However, for those who contend that agency is an all-or-nothing proposition, hinging on complete autonomy, the notion of partial responsibility is in dispute (Hakli & Makela, 2019). For the semi-autonomous robot or AI, government regulation can provide recourse for badly behaving robots, such as the European framework for regulating robotics, “Robolaw,” employing a case study approach, weighing, among other things, questions of liability when humans or their interests are harmed (Palmerini, et al., 2016). The South Korean government's Robot Ethics Charter offers another regulatory framework for establishing parameters around safe robot operation (Kim & Kim, 2013).

The futures-based Regulatory scenario envisions a more pervasive and encompassing regulatory environment in which robots and AI “own” (are responsible for) that portion of the harm that can be attributed to them, while humans own their part. This scenario is perhaps the best and most likely template for future ethical technology development. The premise is that ethical human-machine interactions will require joint coordinated action, a function of combined human and machine inputs, outputs, and their outcomes. Flexible, case-sensitive regulations, such as those proposed by the “Robolaw” framework, can provide guardrails and parameters for ethical behaviors. Together with supportive policies, they can serve as scaffolding for both programmed and machine learning-based algorithms.

When planning for a more desired future, we must first confront its essential unknowable nature. That is, there are any number of possible futures, and the one that emerges may or may not correspond to our expectations. This suggests that in addition to using foresight methods to explore possible futures, we also need to institute a rigorous design and evaluation regimen that will allow course corrections as the situation changes, different scenarios are tested, and new data come in. The future is not only in the making but is also in the re-making, empowering the users to provide feedback on the desirability of the scenarios, thereby supporting the evolution

of the machine-human collaboration. Robot/AI contributions to greater future capabilities, accessibility, and participation for people with disabilities and older adults are predicated upon the evolution of more transparent, accountable, and empowering human-machine interactions and collaborative scenarios. Participatory user-centric design approaches can infuse the values, aspirations, and needs of older adults and people with disabilities into the algorithms, dovetailing with the collaborative approach of the Regulatory scenario. These factors combined make the flexible Regulatory scenario the most desirable moving forward.

Discussion

Policy recommendations

The intelligent AT discussed are emergent in their use and application. Consumers may not have encountered the technology, owing to accessibility issues (i.e., usability, affordability, knowledge), prompting some investigators to call for the development of “anticipatory ethics” that harken to the need for non-technical values, such as information equity, privacy, anonymity, and democratization as the basis for an evolving ethical design (Durocher et al., 2017; Fields et al., 2021; Shilton, 2015). A parallel process is called for in regard to policy development, taking stock of the emergent nature of the technology, and the key role for consumers in its design and implementation. Other researchers have connected participatory and value-sensitive designs to inclusive policy development for underrepresented groups, including older adults and people with disabilities (Baker et al., 2015; Gandy et al., 2017; Young et al., 2019). It is clear that to be sustainable, relevant, and effective policies must reflect the voices and aspirations of the older adults and people with disabilities whose lives they directly affect while taking into account the multiple stakeholders who constitute what Young et al. (2019) term the “complex policymaking ecosystem” in which technology figures prominently (p. 89). Social and cultural factors play a central role in the iterative policy development process that makes technology responsive to the needs and aspirations of older adults and people with disabilities (Baker et al., 2015; Gandy et al., 2017).

A more inclusive future that advances the empowerment of, and with, people with disabilities, requires policies that regulate the design of robots, AI, and intelligent systems within certain parameters. These parameters must be broad enough to extend beyond “safe” or “non-harmful” use, to include optimizing the users’ physical cognitive and decision-making capabilities. In the case of social robots, for example, this includes social skill practice, as well as forming a bond or connection with the robot. The robots must be developed with factors, such as social acceptance in mind, as well as user preferences. Regulations could require that for a robot to serve as a social (or companion) robot it must demonstrably appear and perform in socially acceptable ways. More fundamentally, it is critical to develop legislation that privileges human rights in the context of human-robot interaction, mandating ethical design by force of law (Magrani, 2017). Value-sensitive design for ethical robots would establish norms around fairness,

reliability, security, privacy, data protection, inclusiveness, transparency, and accountability in terms that reflect the user’s priorities (Magrani, 2017). The key is to combine technology design, proactive policy supports, and adaptable interfaces with iterative developmental and evaluative processes that invite inputs and participation from the users with a disability. This approach ensures that a more inclusive future of evolving technologies is constantly informed and shaped by people with a disability.

Practice recommendations

Accessibility, broadly understood to encompass factors such as technical literacy, affordability, and equitable access, leveraged by user-centric design promotes IAT development to reflect user needs and aspirations (Ienca et al., 2018; Santos et al., 2019). Stakeholder groups, such as people with disabilities and older adults, have distinctly different notions of acceptable technologies, further underscoring the need for user-centric designs focused on end-users (Bradwell et al., 2019). Understanding the needs of all potential users has been identified as a key factor in the success of digital inclusion in Europe, influenced by socio-economic factors, in particular, economic disadvantage, and discrimination, creating barriers to AT uptake by people with disabilities and older adults (Whitney et al., 2011).

Education, in the form of courses and materials that inform and raise awareness, provides a tool for greater inclusiveness (Whitney et al., 2011). Design for all, also called universal design, compliments accessibility efforts (Whitney et al., 2011), as does an “ethical by design” approach, bolstered by workshops actively engaging consumers and their caregivers in discussions to enhance design and implementation (Mulvenna et al., 2017). Our approach combines a universal design perspective, together with an emphasis on ethical design, while factoring in the COVID-19 socio-technical context of austerity, greater technology use, and human-robot/AI collaboration. This leads us to suggest expanded learning opportunities for users, families, and providers that include the development of online communities of practice.

Conclusion

The well-documented adverse economic effects of the COVID-19 pandemic (Chui & Ko, 2020), especially for marginalized populations such as people with disabilities, are likely to have persistent effects that will impact the affordability and access of AI, robots, and intelligent assistive technologies, requiring more cost-effective intelligent systems and policies that promote access. Distributed virtual intelligent systems with less costly interfaces are conceivable as one viable response to likely cost and accessibility challenges. The scale and complexity of these networks would likely increase the need for enhanced data management, and data security measures in the face of increased risks of breaches of privacy. Distributed, complex virtual intelligent networks would also make assessments of negligence and/or culpability for the robot(s) and AI within the networked intelligent systems more difficult to apportion, or

perhaps even identify, but greater transparency in IAT design and implementation will assist with this.

The “humane in the loop” urges us to consider how the options presented by our design choices and commitment to an iterative, co-creation process with the end-user, inform the scope of our normative imagination and choice making. How IATs are shaped, implemented, and adapted presents a mirror from which to reflect on the cumulative societal effects of the dance of human and machine abilities. Our perspective privileges human priorities and subjective values, equity, and inclusiveness as essential, a human-centric perspective, but one that acknowledges emergent properties of technology development in environments characterized by pervasive technologies and collective human-machine intelligence (Groth et al., 2018; Peeters et al., 2020). The inflection point ahead is one in which humans and machines, including IAT co-evolve in an environment of “ambient intelligence,” that is, of invisible, intelligent, and intuitive interfaces in places where people live, work and participate socially (Fellbaum, 2007; Moreno et al., 2013; Teles et al., 2019). Ethics have a central place in grounding IAT design and implementation in recognizable human (humane) values respecting and advancing the autonomy and agency of older adults and people with disabilities.

Older adults with physical, cognitive, and/or functional limitations and people with disabilities continue to be marginalized populations, particularly vulnerable to harms that arise from technology development processes that are de-humanizing. This leads us to question the prevailing “functional approach” notion of IAT that would have us prioritize discrete performances, or “doings,” and efficiency above holistic, potentialities, or “qualities of being” and capacity building. In its stead, we propose a participatory, user-centric approach to the design and implementation of IAT grounded in first principles of beneficence, autonomy, agency, and equity. IAT-user enabled progress will be measurable along a continuum that blends functionality with meaningful progress toward valued outcomes, impacts, and learning targets. As we look toward the future, while continuing to grapple with the uncertainty and socio-technical challenges of COVID-19, we propose inclusive policies and practices to pave the way for IAT that serve a humane purpose.

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